

JavaScript Arrays, Strings & Functions

Total Questions: 10

Topics Covered: 1D Arrays, 2D Arrays, Strings, Functions

Instructions: Use functions wherever possible and test your code with multiple inputs.

1. Number Statistics Calculator

Create a function `getStatistics(numbers)`.

Requirements:

- Find the largest number.
- Find the smallest number.
- Calculate the sum.
- Calculate the average.

Input:

[12, 45, 67, 23, 89, 10]

Expected Output:

Largest Number: 89

Smallest Number: 10

Sum: 246

Average: 41

2. Even and Odd Number Analyzer

Create a function that separates even and odd numbers.

Input:

[5, 8, 12, 15, 20, 33]

Expected Output:

Even Numbers: [8, 12, 20]

Odd Numbers: [5, 15, 33]

Even Count: 3

Odd Count: 3

3. Duplicate Element Finder

Find all duplicate values from an array.

Input:

[10, 20, 30, 20, 40, 10, 50]

Expected Output:

Duplicate Values: [10, 20]

Bonus:

Show occurrence count of each duplicate.

4. String Formatter Tool

Perform the following operations on a string:

- Convert to uppercase
- Convert to lowercase
- Count vowels
- Count consonants

Input:

JavaScript

Expected Output:

Uppercase: JAVASCRIPT

Lowercase: javascript

Vowels: 3

Consonants: 7

5. Word Frequency Counter

Count occurrences of each word.

Input:

apple banana apple orange banana apple

Expected Output:

apple : 3

banana : 2

orange : 1

Bonus:

Ignore case sensitivity.

6. Matrix Sum Calculator (2D Array)

Calculate row sums, column sums and total sum.

Input:

```
[  
  [1,2,3],  
  [4,5,6],  
  [7,8,9]  
]
```

Expected Output:

Row Sums: 6, 15, 24

Column Sums: 12, 15, 18

Total Sum: 45

7. Student Result Management System

Input:

```
[  
  ["Rahul", 80],  
  ["Priya", 92],  
  ["Aman", 65],  
  ["Neha", 88]  
]
```

Requirements:

- Find topper
- Find average marks
- Show students above average

Expected Output:

Topper: Priya (92)

Average Marks: 81.25

Above Average Students:

Priya

Neha

8. Password Validator

Rules:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number

Input:

JavaScript123

Expected Output:

Valid Password

Input:

javascript

Expected Output:

Invalid Password

9. Shopping Cart System

Input:

[500, 1200, 300, 800]

Requirements:

- Calculate total bill
- Find most expensive item
- Find cheapest item
- Find average price

Expected Output:

Total Bill: 2800

Most Expensive: 1200

Cheapest: 300

Average Price: 700

10. Employee Management System

Input:

```
[  
  ["E101", "Rahul", 35000],  
  ["E102", "Priya", 42000],  
  ["E103", "Aman", 28000]  
]
```

Requirements:

- Search employee by ID
- Find highest salary
- Find lowest salary
- Calculate average salary
- Display all employee details

Expected Output:

Highest Salary: Priya - 42000
Lowest Salary: Aman - 28000
Average Salary: 35000